

A spectral comparison for worst-orientation Nyström approximation

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Abstract

For a prescribed strictly positive spectrum, let x_k be the largest optimal coordinate Nyström error over its real orthogonal orbit, and let y_k be the determinant-weighted mean error. We prove, for every dimension and selection size,

$$x_k \leq y_k \leq (k+1) \min \left\{ 1, \frac{768 \cdot 10^{27} H_{\min(k, n-k)} + 1}{\sqrt{k}} \right\} x_k,$$

where $H_d = \sum_{j=1}^d j^{-1}$. The proof constructs one orientation that is difficult for every selected coordinate subset. A Gaussian inverse-trace estimate produces a frame robust to free observations and repetitions; a grouping construction then accommodates arbitrary long spectral tails. No spectral-gap, condition-number, or multiplicity assumption is imposed. We also give quantitative bounds for two-level spectra and a factor-eight reduction to logarithmically many positive spectral levels. The result is an asymptotic improvement over the elementary linear comparison, with a very large explicit constant. It does not establish a dimension-independent bound or matching growth.

Contribution and significance

The new comparison fixes *every eigenvalue in advance* and then constructs one orientation that defeats *every k -coordinate selection*. This prescribed-spectrum quantifier is absent from the nearby good-subset guarantees and hard-instance constructions compared in Section 1. The resulting $O(\sqrt{k}[1 + \log \min(k, n-k)])$ factor holds throughout the positive spectral cone. A separate factor-eight reduction preserves the global comparison question while reducing to logarithmically many spectral levels; it does not assert two-level extremality.

The distinctive mechanism makes a single frame survive both kernel compression by freely observed rows and arbitrary observation repetitions. A trace-zero block construction pools a diffuse tail into balanced noise groups while preserving all prescribed eigenvalues. Together these steps avoid choosing an orientation after seeing the selector and remove logarithmic dependence on the long tail length. This spectral preservation is the conceptual value of the argument; its large constants preclude a practical sharpness claim. Classical sampling identities and Gaussian tools serve as background, while the uniform construction and spectral reduction carry the contribution. The dimension-independent tightness question remains open.

1. The problem and the role of the construction

Let $n \geq 2$ and $1 \leq k < n$ be integers. For an ordered spectrum $\lambda_1 \geq \dots \geq \lambda_n > 0$, let

$$O_\lambda = \{V^\top \text{diag}(\lambda)V : V \in O(n)\},$$

where $O(n)$ is the real orthogonal group. For a subset $I \subseteq [n] = \{1, \dots, n\}$ of size k , define the residual trace

$$E_K(I) = \text{tr}(K - K_{:,I}K_{I,I}^{-1}K_{I,:}).$$

Strict positivity makes every inverse in this definition well-defined. Set

$$x_k(\lambda) = \max_{K \in O_\lambda} \min_{|I|=k} E_K(I), \quad y_k(\lambda) = (k+1) \frac{e_{k+1}(\lambda)}{e_k(\lambda)},$$

where $e_j(\lambda)$ is the sum of all products of j distinct entries and $e_0 = 1$. The maximum exists: the orthogonal group is compact, each residual is continuous, and the minimum is over finitely many subsets. We abbreviate

$$m = n - k, \quad s = \sum_{i=k+1}^n \lambda_i, \quad F = \frac{e_{k+1}(\lambda)}{e_k(\lambda)}, \quad H_d = \sum_{j=1}^d \frac{1}{j} \quad (d \geq 1).$$

Theorem 1.1 (Main comparison). *For every $n \geq 2$, $1 \leq k < n$, and strictly positive spectrum λ ,*

$$x_k(\lambda) \leq y_k(\lambda) \leq (k+1) \min \left\{ 1, \frac{768 \cdot 10^{27} H_{\min(k, n-k)} + 1}{\sqrt{k}} \right\} x_k(\lambda).$$

The proof is completed in Section 9. It gives a bound of order $\sqrt{k} [1 + \log \min(k, n-k)]$ with an absolute, explicitly displayed constant. The construction proves existence of a difficult orientation for every prescribed spectrum; it is not an efficient procedure for computing such an orientation. In particular, the large numerical constant cannot be discarded when the estimate is used in a fixed dimension.

The diagonal orientation gives $x_k \geq s$. Determinant-weighted averaging will give $x_k \leq y_k$, and an elementary coefficient estimate gives $y_k \leq (k+1)x_k$. The substantive task is to improve this last factor uniformly over spectra by constructing an orientation whose minimum error is large.

The order of the quantifiers matters. An orientation is chosen before the selected set I . Every construction below therefore controls all k -subsets at once. The orientation may depend on a spectral threshold or a tail prefix. Bounds obtained from different orientations can be maximized; they cannot be added.

We use $A \leq B$ for positive semidefinite matrix order, $\|\cdot\|_2$ for the Euclidean operator norm, and $\|\cdot\|_F$ for the Frobenius norm. A matrix Q with $Q^\top Q = I$ is called a Parseval frame. Its rows are the frame vectors. All matrices and vector spaces are real. The constants retained throughout are

$$D = 10^{12}, \quad C_0 = 4 \cdot 10^{27}, \quad C_1 = 32 \cdot 10^{27}.$$

The appendices also give a strict improvement of the explicit dimensional factor for every fixed interior pair and comparisons without a harmonic loss near arbitrary anisotropic rank boundaries; see Appendices D and E.

1.1. Context and proof structure

Problem 4.7 in the Simons workshop collection [1, Section 4.2] asks about the tightness of the determinant-weighted error for exactly this fixed-spectrum maximum over orientations followed by a minimum over coordinate subsets. It explicitly notes equality when $k = n - 1$. Its formulation is a tightness question; the dimension-independent supremum considered at the end of this paper is one way to quantify that question.

Guruswami and Sinop [2, Eq. (1.2) and Theorem 4.1] identify the determinant-weighted mean and bound it in terms of a best-rank spectral tail. Applied to $K^{1/2}$, their squared Frobenius projection error is the Nyström residual trace used here. Fornace and Lindsey [3, Proposition 5.1 and Corollary 3.1] identify the same benchmark and compare greedy captured trace with it. These statements control selection performance or its mean. They do not provide the prescribed-spectrum lower bound on x_k required by our comparison.

The lower-bound construction of Dereziński, Khanna, and Mahoney [5, Theorem 3 and Appendix F] is closer in direction: it makes every subset at specified sizes incur a large error relative to a best-rank tail. Its scales are chosen in the construction. Our theorem instead takes the complete positive eigenvalue list as input, so that none of its levels can be adjusted. The distinction concerns the prescribed-spectrum quantifier, not positive definiteness: their examples also use positive perturbations.

For Parseval frames, Avron and Boutsidis [4, Theorem 6 and Lemma 14] guarantee a basis with small inverse Frobenius norm. Their Theorem 21 gives a different lower-bound family; at fixed aspect ratio and basis-size selection its normalized lower factor stays bounded. Damle [6, Corollary 2] gives $\|Q_I^{-1}\|_F^2 \leq k(n - k + 1)$ through Bischof–Stewart pivoting. These do not supply the $\Omega(k^{3/2})$ simultaneous lower bound of Appendix A, or the kernel and repetition guarantees in Lemma 5.1.

Kozyrev and Osinsky [7, Theorem 1], and Osinsky and Kozyrev [8, Theorems 1–2], develop further good-subset and exchange bounds. The former explicitly distinguishes worst-case pseudoinverse ratios for arbitrary full-rank matrices from the extremum over matrices with orthonormal rows; its sharp arbitrary-matrix ratio cannot be substituted for the latter. Neither result prescribes an entire covariance spectrum and constructs an orientation that is difficult for every coordinate subset. This comparison locates the contribution of the present proof in the uniform spectral-orbit bound and its supporting simultaneous construction, rather than in the classical sampling identities. All mathematical dependencies are proved below.

The central probabilistic step is a lower-tail estimate for the inverse trace of a rectangular Gaussian Gram matrix. A union bound and whitening turn this into a single frame with simultaneous control of every relevant row subset. Kernel compression handles rows that are already observed, while the same estimate handles repeated observations. A spectral prefix then yields a finite-noise lower bound. When the tail is too diffuse for a short prefix, a balanced grouping construction pools its mass without altering the prescribed eigenvalues. An elementary symmetric-polynomial inequality supplies the final choice of spectral threshold.

The appendices retain additional consequences useful for future work. Appendix A gives a stronger constant for square bases and two-level spectra; Appendix B treats a shifted-rank boundary; Appendix C reduces arbitrary spectra to a bounded range and then to logarithmically many positive levels.

2. Elementary matrix and probability tools

2.1. Inverse order and compression

If $0 < A \leq B$, then $B^{-1} \leq A^{-1}$. Indeed, $A^{-1/2}BA^{-1/2} \geq I$; orthogonal diagonalization and inversion give the assertion after conjugation. If $T > 0$ and U has orthonormal columns, then

$$\mathrm{tr}(T^{-1}) \geq \mathrm{tr}((U^T T U)^{-1}). \quad (2.1)$$

To prove this, complete U to an orthogonal basis and write T in that basis as $\begin{pmatrix} A & B \\ B^T & C \end{pmatrix}$. When the complement has positive dimension, the corresponding diagonal block of T^{-1} is $(A - BC^{-1}B^T)^{-1} \geq A^{-1}$; the remaining diagonal block has nonnegative trace. If the complement is empty, equality holds. The block formula follows by multiplying the two block matrices. In particular, the inverse trace of a principal block cannot decrease when its set of indices is enlarged.

If $J = I^c$, the same block formula identifies the nonzero block of the Nyström residual:

$$E_K(I) = \mathrm{tr}(((K^{-1})_{J,J})^{-1}). \quad (2.2)$$

2.2. Scalar regularization

Suppose $P \geq 0$ and every nonsingular principal block $P_{J,J}$ of a fixed size has inverse trace at least $L > 0$. If $T > 0$ satisfies $T \leq \alpha I + \beta P$ with $\alpha, \beta > 0$, then every block of that size satisfies

$$\mathrm{tr}(T_{J,J}^{-1}) \geq \frac{L}{\beta + \alpha L}. \quad (2.3)$$

For a nonsingular block let $z_i > 0$ be its inverse eigenvalues and put $Z = \sum_i z_i \geq L$. Inverse order gives

$$\mathrm{tr}(T_{J,J}^{-1}) \geq \sum_i \frac{z_i}{\beta + \alpha z_i} \geq \frac{Z}{\beta + \alpha Z} \geq \frac{L}{\beta + \alpha L}.$$

The middle inequality uses $z_i \leq Z$; the last uses monotonicity in Z . If $P_{J,J}$ is singular, $(\alpha I + \beta P_{J,J})^{-1}$ has an eigenvalue $1/\alpha$, already larger than the required bound. This is why later zero rows and repeated rows cause no missing selection cases.

2.3. Gaussian squares and finite nets

A standard Gaussian has density $(2\pi)^{-1/2} \exp(-u^2/2)$. If X is a sum of d independent Gaussian squares, direct integration gives

$$\mathbb{E}e^{zX} = (1 - 2z)^{-d/2}, \quad z < \frac{1}{2}. \quad (2.4)$$

The joint density depends only on the Euclidean norm. Orthogonal coordinate changes therefore preserve the standard Gaussian distribution and coordinate independence. In particular, conditionally on a subspace determined by some rows, independent new rows have independent standard Gaussian coordinates after projection onto any orthonormal basis of that subspace.

A nonzero polynomial in finitely many independent Gaussians vanishes with probability zero. Induct on the number of variables, expand in the last one, and outside the zero set of a nonzero coefficient polynomial use the finiteness of the roots of a nonzero univariate polynomial. Gaussian Gram determinants of the relevant rectangular sizes are nonzero polynomials, as witnessed by a

rectangular identity matrix. Thus all finitely many maximal-rank requirements imposed below hold simultaneously almost surely.

For $0 < \eta < 1$, the unit sphere of $\mathbb{R}^d$ has an η -net of at most $(1 + 2/\eta)^d$ points. A maximal set of points at pairwise distance greater than η has this property: disjoint balls of radius $\eta/2$ around the points lie in the ball of radius $1 + \eta/2$, giving the cardinality bound by volume comparison. That bound also ensures a finite maximal set exists. For any matrix Z , approximation of a maximizing unit vector gives

$$\|Z\|_2 \leq (1 - \eta)^{-1} \max_{v \text{ in the net}} \|Zv\|. \quad (2.5)$$

If $\|Z\|_2 \leq M$ and $\|Zv\| \geq b$ on the net, the triangle inequality further gives $\|Zx\| \geq b - \eta M$ for every unit vector x . These estimates will be used only for existence, with explicit finite union bounds.

3. A Gaussian inverse-trace lower tail

Lemma 3.1 (Gaussian inverse-trace lower tail). *Let $t \geq 1$, $q \geq 0$, and $p = t + q \geq 4$ be integers. If G is a $t \times p$ matrix of independent standard Gaussians, then*

$$\mathbb{P} \left\{ \text{tr}((GG^\top)^{-1}) \leq \frac{t}{1000(q + \sqrt{p})} \right\} \leq 2e^{-31p}. \quad (3.1)$$

Proof. Put $A = (GG^\top)^{-1}$ and $j = \min(t, \lfloor \sqrt{p} \rfloor)$. On the event in (3.1), the average of $\sum_{i \in S} A_{ii}$ over j -subsets S of the row indices is at most $j/[1000(q + \sqrt{p})]$. At least half those subsets consequently have sum at most $j/[500(q + \sqrt{p})]$, by Markov's inequality. Row permutation invariance gives

$$\mathbb{P} \left\{ \text{tr} A \leq \frac{t}{1000(q + \sqrt{p})} \right\} \leq 2\mathbb{P} \left\{ \sum_{i=1}^j A_{ii} \leq \frac{j}{500(q + \sqrt{p})} \right\}. \quad (3.2)$$

Write the first j rows of G as U and the remaining rows as B . Conditionally on B , choose an orthonormal basis matrix R for $\ker B$, of dimension $q + j$. When $t = j$ this is the whole ambient space. The inverse block identity gives

$$A_{[j],[j]} = (URR^\top U^\top)^{-1}.$$

For $t > j$, this follows from $RR^\top = I - B^\top(BB^\top)^{-1}B$; for $t = j$, it follows directly from the definition of A . Conditionally on B , the matrix $W = UR$ is $j \times (q + j)$ standard Gaussian.

If $\sigma_1, \dots, \sigma_j$ are its positive singular values, then $(\sum_i \sigma_i^2)(\sum_i \sigma_i^{-2}) \geq j^2$ by Cauchy–Schwarz. The event on the right of (3.2) therefore implies

$$\|W\|_F^2 \geq 500j(q + \sqrt{p}) \geq 500j(q + j).$$

There are $d' = j(q + j)$ independent squared Gaussian entries in this norm. Equation (2.4) at $z = 1/4$ bounds the conditional probability by

$$2^{d'/2} e^{-125d'} \leq e^{-124d'}.$$

If $t \geq \lfloor \sqrt{p} \rfloor$, then $j \geq \sqrt{p}/2$ and $d' \geq p/4$. Otherwise $j = t$ and $d' = t(q + t) = tp \geq p$. The bound e^{-31p} thus holds in both cases, also after removing the conditioning. Equation (3.2) proves the lemma. No independence between inverse diagonal entries was used. $\square$

4. Projections that control every principal subset

Lemma 4.1 (Uniform principal-block control). *For integers $t \geq 1$, $q \geq 0$, $p = t + q$, and $N \geq \lceil 4p/3 \rceil$, there is a rank- p orthogonal projection P on $\mathbb{R}^N$ such that every nonsingular t -principal block satisfies*

$$\mathrm{tr}(P_{J,J}^{-1}) \geq \frac{Nt}{C_0(q + \sqrt{p})}. \quad (4.1)$$

For this same P , if $T > 0$ and $T \leq \alpha I + \beta P$ with $\alpha, \beta > 0$, every t -subset, including those with singular $P_{J,J}$, satisfies (2.3) with $L = Nt/[C_0(q + \sqrt{p})]$.

Proof. First assume $p \geq 4$ and $4p/3 \leq N_0 \leq 2p$. Let Z be an $N_0 \times p$ standard Gaussian matrix. For each fixed unit vector v , $\|Zv\|^2$ is a sum of N_0 Gaussian squares. A $1/2$ -net, (2.4), and (2.5) give

$$\mathbb{P}\{\|Z\|_2 > 10\sqrt{p}\} \leq 5^p 2^{N_0/2} e^{-25p/4} \leq e^{(\log 10 - 25/4)p} < e^{-3p}. \quad (4.2)$$

For the lower singular value use an η -net with $\eta = 1/(20D)$, of size at most $(41D)^p$. At each net point, (2.4) at $z = -D^2/4$ gives

$$\mathbb{P}\{\|Zv\|^2 \leq 4p/D^2\} \leq e^p (1 + D^2/2)^{-N_0/2} \leq e^p (2/D^2)^{2p/3}.$$

The probability that some net point fails this bound is at most

$$\left(\frac{41e \cdot 2^{2/3}}{D^{1/3}}\right)^p < \left(\frac{246}{10000}\right)^p < 16^{-p}. \quad (4.3)$$

Here $e < 3$ and $2^{2/3} < 2$ suffice. Outside the events in (4.2)–(4.3), every unit vector x satisfies

$$\|Zx\| \geq \frac{2\sqrt{p}}{D} - \frac{10\sqrt{p}}{20D} \geq \frac{\sqrt{p}}{D}, \quad \lambda_{\min}(Z^\top Z) \geq p/D^2. \quad (4.4)$$

There are at most $2^{N_0} \leq 4^p$ row subsets of size t . Lemma 3.1 and a union bound give

$$\mathbb{P}\left\{\exists J : \mathrm{tr}((Z_J Z_J^\top)^{-1}) \leq \frac{t}{1000(q + \sqrt{p})}\right\} \leq 2 \cdot 4^p e^{-31p} \leq 2e^{-29p}. \quad (4.5)$$

The sum $e^{-3p} + 16^{-p} + 2e^{-29p}$ is less than one for $p \geq 4$. For example, $e > 2$ bounds the terms by 2^{-12} , 2^{-16} , and 2^{-115} . Hence a realization exists satisfying (4.4), the complement of (4.5), and all required ranks.

Fix it and put $V = Z(Z^\top Z)^{-1/2}$, so $V^\top V = I_p$. By inverse order and (4.4),

$$(VV^\top)_{J,J} \leq (D^2/p) Z_J Z_J^\top, \quad \mathrm{tr}((V_J V_J^\top)^{-1}) \geq \frac{pt}{10^{27}(q + \sqrt{p})}. \quad (4.6)$$

Now take $N_0 = \min(N, 2p)$. If $p \leq 3$, instead use the matrix whose first p rows are I_p and whose other rows are zero. Every nonsingular block then has inverse trace t , which is at least the right side of (4.6). Thus (4.6) is available for all $p \geq 1$.

Let $h = \lfloor N/N_0 \rfloor$. Replace each row of V by h copies divided by $\sqrt{h}$, and append $N - hN_0$ zero rows. The resulting $N \times p$ matrix V' still has orthonormal columns. A nonsingular t -row Gram matrix uses neither a zero row nor two copies of the same original row. Its inverse trace is h times the corresponding inverse trace in (4.6). Since

$$h \geq \frac{N}{2N_0} \geq \frac{N}{4p},$$

$P = V'V'^T$ satisfies (4.1). The final assertion is exactly scalar regularization (2.3). $\square$

5. One frame for free rows and repeated observations

Replication is useful only if observations of the same original row are controlled. The next lemma gives that control and the stronger kernel statement needed for spectral grouping. Observing a set of rows for free means passing to their common kernel, where those observations contain no signal; the remaining rows must still have a large inverse trace. Both assertions below hold for one and the same frame.

Lemma 5.1 (A frame robust to free rows and repetitions). *Let $k \geq 1$ and $0 \leq q \leq \lfloor k/2 \rfloor$, and put $r = k - q$. There is a $2k \times r$ Parseval frame Q_0 with every set of at most r rows independent, having both of the following properties.*

For every pair of disjoint row sets $H, L \subseteq [2k]$ with $h = |H| \leq \lfloor k/4 \rfloor$ and $h + |L| \leq k$, and every orthonormal basis matrix U for $\ker((Q_0)_H)$, put $G = (Q_0)_L U$. Whenever G has full column rank,

$$\text{tr}((G^T G)^{-1}) \geq \frac{k^2}{4 \cdot 10^{27}(q + \sqrt{k})}. \quad (5.1)$$

For every nonnegative integer tuple $w = (w_1, \dots, w_{2k})$ with $\sum_i w_i = k$, whenever $Q_0^T \text{diag}(w) Q_0$ is nonsingular,

$$\text{tr}((Q_0^T \text{diag}(w) Q_0)^{-1}) \geq \frac{k^2}{8 \cdot 10^{27}(q + \sqrt{k})}. \quad (5.2)$$

Proof. Suppose first $k \geq 16$ and let X be $2k \times r$ standard Gaussian. Fix eligible disjoint H, L , write $l = |L|$, and condition on X_H . Its row rank is h almost surely. If $l < r - h$, the compressed matrix cannot have full column rank and imposes no requirement. Otherwise put $r' = r - h \geq k/4$ and choose an orthonormal basis matrix U_0 for $\ker X_H$. The matrix $X_L U_0$ is conditionally standard Gaussian of size $l \times r'$. Its transpose has r' rows and $p' = l$ columns, with

$$0 \leq q' = l - r' = l + h - r \leq q, \quad k/4 \leq r' \leq p' \leq k.$$

In particular $p' \geq 4$, and

$$\frac{k}{4000(q + \sqrt{k})} \leq \frac{r'}{1000(q' + \sqrt{p'})}.$$

Lemma 3.1 applied to this transpose bounds the probability of failure of the inverse-trace lower bound $k/[4000(q + \sqrt{k})]$ by $2e^{-31k/4}$. The bound is uniform in the conditioned rows, so also holds unconditionally. There are at most $3^{2k} = 9^k$ disjoint pairs H, L . Therefore the probability that any eligible pair fails is at most

$$2 \cdot 9^k e^{-31k/4} \leq 2e^{-4k}. \quad (5.3)$$

We also need the full matrix to be well-conditioned. The same net estimates as in Section 4, now with $2k$ rows and $r \leq k$ columns, give

$$\mathbb{P}\{\|X\|_2 > 10\sqrt{k}\} \leq 5^k 2^k e^{-25k/4} < e^{-3k}.$$

At a fixed point of an $\eta = 1/(20D)$ net, $\mathbb{P}\{\|Xv\|^2 \leq 4k/D^2\} \leq e^k (2/D^2)^k$. Multiplying by the net size at most $(41D)^k$ bounds the failure probability by $(82e/D)^k < 16^{-k}$. The triangle inequality on that net then gives

$$S = X^\top X \geq (k/D^2)I_r \quad (5.4)$$

outside a set of probability at most $e^{-3k} + 16^{-k}$. Together with (5.3), the total failure probability is less than one for $k \geq 16$; using $e > 2$ bounds it by $2^{-48} + 2^{-64} + 2^{-63}$. Choose a realization satisfying all these requirements and independence of every set of at most r rows. Set $Q_0 = XS^{-1/2}$.

It remains to check that whitening preserves the kernel estimates. For fixed H , let $B_H = U_0^\top S U_0$ and define

$$U = S^{1/2} U_0 B_H^{-1/2}.$$

Then $U^\top U = I_{r-h}$ and $(Q_0)_H U = 0$, so its columns form an orthonormal basis of the required kernel. Moreover

$$(Q_0)_L U = X_L U_0 B_H^{-1/2}.$$

Consequently, by (5.4) and the chosen inverse-trace bound,

$$\begin{aligned} \text{tr}(((Q_0)_L U)^\top (Q_0)_L U)^{-1}) &= \text{tr}(B_H((X_L U_0)^\top X_L U_0)^{-1}) \\ &\geq \frac{k}{D^2} \frac{k}{4000(q + \sqrt{k})} = \frac{k^2}{4 \cdot 10^{27}(q + \sqrt{k})}. \end{aligned}$$

The trace is invariant under orthogonal changes of the kernel basis. This proves (5.1) for every allowed U , using the single chosen frame.

For $1 \leq k < 16$, choose any Parseval frame with every set of at most r rows independent. Gaussian construction followed by whitening provides one by the polynomial argument of Section 2. Since $(Q_0)_L^\top (Q_0)_L \leq I_r$, every nonsingular $G^\top G$ in (5.1) is at most I_{r-h} . Its inverse trace is at least $r - h \geq k/4$, whereas the right side of (5.1) is at most $k^{3/2}/(4 \cdot 10^{27}) < 1/4$. Thus these dimensions are covered as well.

To deduce (5.2), write $B = Q_0^\top \text{diag}(w) Q_0 > 0$ and define $s_0 = |\{i : w_i > 0\}|$, the support size of the tuple w . Then $s_0 \geq r$. Let H consist of indices with $w_i \geq 3$, and let L be the rest of the support. Since

$$2|H| \leq \sum_{i:w_i > 0} (w_i - 1) = k - s_0 \leq q,$$

we have $h = |H| \leq k/4$ and $h + |L| = s_0 \leq k$. For an orthonormal basis U of $\ker((Q_0)_H)$,

$$0 < U^\top B U \leq 2G^\top G, \quad G = (Q_0)_L U.$$

Hence G has full column rank. Inverse compression (2.1), inverse order, and (5.1) yield

$$\text{tr}(B^{-1}) \geq \text{tr}((U^\top B U)^{-1}) \geq \frac{1}{2} \text{tr}((G^\top G)^{-1}),$$

which proves (5.2) for all repetitions, with no bound on an individual count w_i . $\square$

6. Every tail prefix at positive noise

Lemma 6.1 (Prefix projection). *For integers $k \geq 1$, $0 \leq q \leq \lfloor k/2 \rfloor$, and $t \geq 1$, there is a rank- $(t+q)$ projection P of size $N = k+t$ whose every nonsingular t -principal block satisfies*

$$\mathrm{tr}(P_{J,J}^{-1}) \geq L_t, \quad L_t = \frac{kt}{C_1(q + \sqrt{k})}. \quad (6.1)$$

For every $T > 0$ with $T \leq \alpha I + \beta P$, the bound $\mathrm{tr}(T_{J,J}^{-1}) \geq L_t/(\beta + \alpha L_t)$ holds for all t -subsets.

Proof. If $t \leq k$, then $3(k+t) - 4(t+q) = 3k - t - 4q \geq 0$. Apply Lemma 4.1 with $p = t+q$. For integral $q \geq 1$, $q + \sqrt{q+t} \leq q + \sqrt{q} + \sqrt{t} \leq 2(q + \sqrt{k})$; the same last inequality holds for $q = 0$. Since $N \geq k$ and $C_1 \geq 2C_0$, (4.1) implies (6.1).

If $t \geq k$, put $r = k - q$ and use the single frame Q_0 from Lemma 5.1. With $h = \lfloor N/(2k) \rfloor \geq N/(4k)$, repeat every row h times after division by $\sqrt{h}$, and append $N - 2kh$ zero rows. This gives a Parseval frame Q of size $N \times r$. For any k -subset I ,

$$Q_I^\top Q_I = h^{-1} Q_0^\top \mathrm{diag}(w) Q_0, \quad w_i \in \mathbb{Z}_{\geq 0}, \quad \sum_i w_i \leq k.$$

If this Gram matrix is nonsingular, add counts to one row type until the total is exactly k . This increases the Gram matrix, so inverse order and (5.2) imply

$$\mathrm{tr}((Q_I^\top Q_I)^{-1}) \geq \frac{hk^2}{8 \cdot 10^{27}(q + \sqrt{k})} \geq \frac{Nk}{C_1(q + \sqrt{k})}. \quad (6.2)$$

Set $P = I_N - QQ^\top$. For complementary sets I, J of sizes k, t ,

$$P_{J,J} = I_t - Q_J Q_J^\top, \quad Q_I^\top Q_I = I_r - Q_J^\top Q_J.$$

Because $t \geq k \geq r$, a singular value decomposition of Q_J shows that the first matrix has the eigenvalues of the second and $t-r$ additional eigenvalues equal to one. They are invertible together, and then

$$\mathrm{tr}(P_{J,J}^{-1}) = t - r + \mathrm{tr}((Q_I^\top Q_I)^{-1}) \geq L_t,$$

using (6.2) and $N \geq t$. The regularized assertion, including every singular auxiliary block, follows from (2.3). $\square$

Proposition 6.2 (A finite-noise prefix bound). *For the spectrum of Section 1 and every $0 \leq q \leq \lfloor k/2 \rfloor$, set*

$$a_q = \lambda_{k-q}, \quad U_q = \frac{ks}{C_1(q + \sqrt{k})H_m}.$$

Then

$$x_k(\lambda) \geq \max \left\{ s, \frac{a_q U_q}{a_q + U_q} \right\}. \quad (6.3)$$

Proof. Fix q and a prefix length $1 \leq t \leq m$, and write $a = a_q$, $b = \lambda_{k+t}$. Use P from Lemma 6.1. Its orthogonal complement has dimension $k-q$; assign the eigenvalues $\lambda_1, \dots, \lambda_{k-q}$ there. Assign $\lambda_{k-q+1}, \dots, \lambda_{k+t}$ to its range. The resulting positive covariance block K_0 has size $k+t$ and precision

$$T_0 = K_0^{-1} \leq a^{-1}I + b^{-1}P.$$

Indeed the precision eigenvalues on the complement are at most a^{-1} , and those on the range are at most b^{-1} . Lemma 6.1 gives, for every t -subset J within this block,

$$\text{tr}((T_0)_{J,J}^{-1}) \geq \frac{abL_t}{a + bL_t}. \quad (6.4)$$

Place the remaining $n - k - t$ covariance eigenvalues in singleton blocks. This forms a positive definite K with the full prescribed spectrum. Any selected k -subset leaves at least t coordinates unselected in K_0 . By (2.2), its error is the inverse trace of the complementary principal precision block. By (2.1), this trace is at least that for any t of the unselected coordinates in K_0 . Thus (6.4) holds for every original selection, regardless of which singleton coordinates are selected. Taking the orbit maximum gives

$$x_k(\lambda) \geq \frac{a\lambda_{k+t}L_t}{a + \lambda_{k+t}L_t} \quad \text{for each } t. \quad (6.5)$$

Let $Z = \max_{1 \leq t \leq m} t\lambda_{k+t}$. Since $\lambda_{k+t} \leq Z/t$, we have $s \leq H_m Z$. Choose a prefix attaining Z . For that one prefix,

$$\lambda_{k+t}L_t = \frac{kZ}{C_1(q + \sqrt{k})} \geq U_q.$$

The function $z \mapsto az/(a + z)$ increases on positive arguments, so (6.5) proves the second bound in (6.3). The first is the diagonal bound $x_k \geq s$. The proof uses one prefix and one orientation for the chosen q , not a sum of errors from different orientations. $\square$

7. Grouping a diffuse long tail

The loss H_m in (6.3) can be replaced by $2H_{8k}$ at the cost of restricting q to $q \leq k/4$. This prevents a long spectral tail from introducing a logarithm of the aspect ratio m/k .

Proposition 7.1 (A diffuse-tail bound). *For $0 \leq q \leq \lfloor k/4 \rfloor$, set*

$$V_q = \frac{ks}{2C_1(q + \sqrt{k})H_{8k}}.$$

Then

$$x_k(\lambda) \geq \max \left\{ s, \frac{a_q V_q}{a_q + V_q} \right\}. \quad (7.1)$$

Proof. Write $r = k - q$ and $a = a_q$. If

$$\sum_{t=1}^{\min(m, 8k)} \lambda_{k+t} \geq s/2, \quad (7.2)$$

restrict the prefix maximization in the proof of Proposition 6.2 to these indices. Its maximum Z satisfies $s/2 \leq H_{8k}Z$. Equation (6.5) then gives (7.1). This case includes every $m \leq 8k$.

Suppose (7.2) fails. In particular $m > 8k$. The positive covariance weights outside the leading r -dimensional space form the list

$$\mathcal{B} = (\lambda_{k-q+1}, \dots, \lambda_n)$$

of length $m + q$. Set aside its first $q + 8k$ entries as singleton coordinates. There remain $p' = m - 8k$ weights of total $S' > s/2$, each less than $w_* = s/(16k)$: any such entry is at most the average of the first $8k$ true tail entries, whose sum is less than $s/2$. Thus $p'w_* > S' > s/2$, so $p' > 8k$.

Reserve the smallest $k+q$ remaining weights. Their sum is at most $(k+q)S'/p' < S'/4$, because the mean of the smallest entries is no larger than the mean of the whole list, $k+q \leq 5k/4$, and $p' > 8k$. The unreserved weights have total $T' > 3S'/4 > 3s/8$, and their number exceeds $2k$.

Partition these weights into $2k$ groups by successively placing the next positive weight into a group of least current total. All groups become nonempty. At each step the largest and smallest totals differ by at most w_* : after increasing a minimum, every total remains between the old minimum and that minimum plus w_* . Consequently the final totals $v_1, \dots, v_{2k}$ satisfy

$$v_{\min} := \min_j v_j \geq \frac{T'}{2k} - w_* > \frac{s}{8k}. \quad (7.3)$$

We next realize each group as a covariance block with a controlled diagonal. If its positive weights are $d_1, \dots, d_p$ of total v , the matrix $\text{diag}(-v, d_1, \dots, d_p)$ is traceless. Every real symmetric traceless matrix admits an orthonormal basis in which its diagonal is zero. To prove this, if the matrix is nonzero it has a positive and a negative eigenvalue. A suitable unit combination of their eigenvectors has Rayleigh quotient zero. Compress to its orthogonal complement, which is again traceless, and induct. The zero matrix and dimension one initialize the induction.

In such a basis write the conjugated group matrix as $D_j - v_j u_j u_j^\top$, where u_j is the unit eigenvector for the negative eigenvalue and D_j retains the positive eigenvalues. Then

$$D_j \geq 0, \quad \ker D_j = \text{span}(u_j), \quad (D_j)_{ii} = v_j (u_j)_i^2. \quad (7.4)$$

Every entry of u_j is nonzero. If $(u_j)_i = 0$, then $(D_j)_{ii} = 0$ and positive semidefiniteness implies $D_j e_i = 0$; for example, diagonalize D_j and use $e_i^\top D_j e_i = 0$. This would make e_i proportional to u_j , contradicting $(u_j)_i = 0$.

Let U_g have the vectors u_j as its columns, supported on disjoint groups. It is an isometry. Choose the frame Q_0 from Lemma 5.1 and complete it to an orthogonal matrix $[Q_0 \ R_0]$, where R_0 has $k+q$ columns. Put the reserved weights in a positive diagonal matrix Δ and define on the group coordinates

$$Q' = U_g Q_0, \quad C' = \text{blockdiag}(D_1, \dots, D_{2k}) + U_g R_0 \Delta R_0^\top U_g^\top. \quad (7.5)$$

The two summands of C' act on orthogonal subspaces. The first has the unreserved positive eigenvalues and kernel $\text{range}(U_g)$. On this kernel the second has the reserved eigenvalues and kernel $\text{range}(Q')$. Hence C' has precisely the p' remaining positive weights and kernel $\text{range}(Q')$.

The number of group coordinates is

$$[p' - (k+q)] + 2k = m - 7k - q.$$

Append the $q + 8k$ singleton coordinates set aside earlier, extending Q' by zero rows to Q and C' by those positive scalar weights to C . The total dimension is $m + k = n$; $Q^\top Q = I_r$; and C has positive spectrum $\mathcal{B}$ and kernel $\text{range}(Q)$. Finally define

$$K = Q A Q^\top + C, \quad A = \text{diag}(\lambda_1, \dots, \lambda_r) \geq a I_r. \quad (7.6)$$

This is a strictly positive definite covariance matrix of the prescribed spectrum. Every choice in (7.3)–(7.6) has been made before any selected set is considered.

Fix now an arbitrary k -subset I , and let E_I be the $k \times n$ matrix selecting its coordinates, so $Q_I = E_I Q$. For predictors $L \in \mathbb{R}^{n \times k}$, completing the square gives

$$E_K(I) = \min_L \operatorname{tr}((I_n - LE_I)K(I_n - LE_I)^\top). \quad (7.7)$$

Indeed the minimizer is $L_* = K_{:,I}K_{I,I}^{-1}$, and the difference from its value is $\operatorname{tr}((L - L_*)K_{I,I}(L - L_*)^\top) \geq 0$. Project the error onto the orthonormal columns of Q and write $M = Q^\top L$. Using $Q^\top C = 0$ and $A \geq aI_r$ gives, for every L ,

$$\operatorname{tr}((I_n - LE_I)K(I_n - LE_I)^\top) \geq a\|I_r - MQ_I\|_F^2 + \operatorname{tr}(MC_{I,I}M^\top). \quad (7.8)$$

Projection decreases the trace of a positive semidefinite matrix, as is seen by completing an orthonormal basis and dropping nonnegative diagonal entries. Expansion of the signal and noise parts in (7.6) gives the right side of (7.8); the terms in the noise expansion not involving L vanish because $Q^\top C = 0$.

If Q_I has rank less than r , a unit vector in its kernel shows $\|I_r - MQ_I\|_F^2 \geq 1$. Thus (7.8) is at least a , sufficient for (7.1). Suppose instead Q_I has full column rank. Let g be the number of selected groups, ignoring singleton coordinates. Each group contributes multiples of one row of Q_0 , so $g \geq r$. Call a group heavy if at least two of its coordinates are selected. Let H be the heavy group indices and L_0 the other selected group indices. If $h = |H|$, counting selected coordinates gives

$$g + h \leq k, \quad h \leq k - g \leq q \leq k/4, \quad h + |L_0| = g \leq k. \quad (7.9)$$

Let U_H be an orthonormal basis matrix for $\ker((Q_0)_H)$, of dimension $r - h$, using the row independence in Lemma 5.1. For each selected light group let u_i be its selected, nonzero entry of u_j , and put these entries in the diagonal matrix D_u . The light rows of Q equal $D_u(Q_0)_{L_0}$. The heavy rows of $Q_I U_H$ vanish, as do singleton rows.

By disjoint group supports, the principal block of $\operatorname{blockdiag}(D_j)$ on the light coordinates consists exactly of the scalar blocks $v_j u_i^2$. Its heavy-coordinate blocks are positive semidefinite. The other summands in C are also positive semidefinite. Therefore

$$C_{I,I} \geq \operatorname{diag}(v_j u_i^2 \text{ on light coordinates; } 0 \text{ elsewhere}). \quad (7.10)$$

This uses the group block structure; it is not an assertion that arbitrary off-diagonal entries of a positive semidefinite matrix may be discarded.

Compress the signal in (7.8) on both sides by U_H , and the noise on its output side by U_H . With

$$T = U_H^\top M_{:, \text{light}} D_u, \quad G = (Q_0)_{L_0} U_H,$$

equation (7.10) gives the lower bound

$$a\|I_{r-h} - TG\|_F^2 + v_{\min}\|T\|_F^2. \quad (7.11)$$

For the signal, all heavy and singleton rows of $Q_I U_H$ are zero. For the noise, apply (7.10) column by column and use $v_j \geq v_{\min}$. Multiplication by an isometry or its transpose decreases a Frobenius norm, because completing the isometry to an orthogonal basis expresses it as omission of nonnegative squares.

The minimum of (7.11) over all matrices T is

$$\text{tr} \left((a^{-1}I_{r-h} + v_{\min}^{-1}G^T G)^{-1} \right). \quad (7.12)$$

A singular value decomposition reduces the minimization to independent scalar problems $\min_z a(1 - z\sigma)^2 + v_{\min}z^2 = av_{\min}/(v_{\min} + a\sigma^2)$, with value a for a zero singular value. Thus (7.12) remains valid when G is singular. Minimizing over all T , even if only some arise from predictors, still provides a lower bound valid for every predictor.

If G is rank deficient, (7.12) is at least a . Otherwise (5.1), (7.3), and (7.9) imply that the inverse trace of $v_{\min}^{-1}G^T G$ is at least

$$v_{\min} \frac{k^2}{4 \cdot 10^{27}(q + \sqrt{k})} \geq Z_0 := \frac{ks}{C_1(q + \sqrt{k})}. \quad (7.13)$$

If its inverse eigenvalues are $z_i > 0$ of sum $Z \geq Z_0$, then

$$\sum_i \frac{z_i}{1 + z_i/a} \geq \frac{Z}{1 + Z/a} \geq \frac{aZ_0}{a + Z_0}.$$

Since $Z_0 \geq V_q$, every predictor error, and hence its minimum (7.7), is at least $aV_q/(a + V_q)$. The same fixed K covers every I . Along with $x_k \geq s$, this proves (7.1) in the diffuse case and completes the proof. $\square$

8. The symmetric quotient and determinant-weighted error

Lemma 8.1 (Elementary symmetric quotient). *With F and s as in Section 1,*

$$F \leq s, \quad \sum_{i=1}^k \frac{F}{\lambda_i + F} \leq \frac{s}{F}. \quad (8.1)$$

Proof. We first prove the coefficient inequality needed for the second assertion. For any finite positive list, let c_j be its elementary symmetric coefficients, extended by zero outside their support. Then

$$c_j^2 \geq c_{j-1}c_{j+1}. \quad (8.2)$$

Induct on the list length, starting with $c_0 = 1$ and all other coefficients zero. Appending $z > 0$ replaces c_j by $d_j = c_j + zc_{j-1}$, and expansion gives

$$\begin{aligned} d_j^2 - d_{j-1}d_{j+1} &= (c_j^2 - c_{j-1}c_{j+1}) \\ &\quad + z(c_jc_{j-1} - c_{j-2}c_{j+1}) \\ &\quad + z^2(c_{j-1}^2 - c_{j-2}c_j). \end{aligned}$$

The first and third terms are nonnegative by induction. On the positive support, (8.2) makes consecutive ratios c_j/c_{j-1} nonincreasing, which implies $c_jc_{j-1} \geq c_{j-2}c_{j+1}$. If the right side is nonzero, all needed indices are in the positive support; otherwise this inequality follows from nonnegativity. This proves the induction, including the boundary indices.

Fix i , put $z = \lambda_i$, and let $A = e_k(\lambda \text{ with } i \text{ omitted})$, $B = e_{k+1}(\lambda \text{ with } i \text{ omitted})$, and $D_* = e_{k-1}(\lambda \text{ with } i \text{ omitted})$. Since $k < n$, $A, D_* > 0$, while $B \geq 0$, and (8.2) gives $A^2 \geq BD_*$. Choose a k -subset with probability proportional to the product of its entries. Its inclusion probability for i is

$$p_i = \frac{zD_*}{A + zD_*}, \quad F = \frac{B + zA}{A + zD_*}.$$

The inequality $z/(z + F) \geq p_i$ follows by multiplying positive denominators and using

$$(A + zD_*)^2 - D_*(B + 2zA + z^2D_*) = A^2 - BD_* \geq 0.$$

Each chosen subset has k entries, so $\sum_i p_i = k$. Hence

$$\sum_{i=1}^n \frac{\lambda_i}{\lambda_i + F} \geq k, \quad \sum_{i=1}^k \frac{F}{\lambda_i + F} \leq \sum_{i=k+1}^n \frac{\lambda_i}{\lambda_i + F} \leq \frac{s}{F}.$$

Finally, every product contributing to $e_{k+1}(\lambda)$ contains a tail entry. Counting such a product at least once gives

$$e_{k+1}(\lambda) \leq \sum_{i=k+1}^n \lambda_i e_k(\lambda \text{ with } i \text{ omitted}) \leq s e_k(\lambda),$$

so $F \leq s$. □

Lemma 8.2 (Determinant-weighted comparison). *For every strictly positive spectrum,*

$$s \leq x_k(\lambda) \leq y_k(\lambda) \leq (k + 1)x_k(\lambda). \quad (8.3)$$

Proof. The first inequality follows from the diagonal orientation. For a positive definite K , the Schur determinant identity for each $j \notin I$ gives

$$\det(K_{I,I})E_K(I) = \sum_{j \notin I} \det(K_{I \cup \{j\}, I \cup \{j\}}). \quad (8.4)$$

Indeed each quotient of determinants on the right is the corresponding residual diagonal entry. Summing (8.4) over k -subsets counts each $(k + 1)$ -principal minor $k + 1$ times. Expansion of $\det(zI + K)$ by selecting diagonal factors z shows that the sum of its principal j -minors is $e_j(\lambda)$. Thus the positive weights $\det(K_{I,I})/e_k(\lambda)$ sum to one and give weighted mean residual $y_k(\lambda)$. The minimum is at most this mean for every orientation; taking the maximum proves $x_k \leq y_k$. Finally $y_k = (k + 1)F \leq (k + 1)s \leq (k + 1)x_k$ by Lemma 8.1. □

9. Proof of the spectral comparison

Theorem 9.1 (Uniform spectral comparison). *Let $n \geq 2$ and $1 \leq k < n$ be integers, and let $\lambda \in (0, \infty)^n$. For*

$$x_k(\lambda) = \max_{V \in O(n)} \min_{\substack{I \subseteq [n] \\ |I|=k}} \text{tr}(K - K_{:,I} K_{I,I}^{-1} K_{I,:}),$$

$$K = V^T \text{diag}(\lambda)V, \quad y_k(\lambda) = (k + 1)e_{k+1}(\lambda)/e_k(\lambda), \quad r_* = \min(k, n - k),$$

where $O(n)$ is the real orthogonal group and e_j is the degree- j elementary symmetric polynomial, one has

$$x_k(\lambda) \leq y_k(\lambda) \leq (k + 1) \min \left\{ 1, \frac{768 \cdot 10^{27} H_{r_*} + 1}{\sqrt{k}} \right\} x_k(\lambda). \quad (9.1)$$

Here $H_d = \sum_{j=1}^d 1/j$. More precisely, with $m = n - k$,

$$\frac{y_k}{x_k} \leq (k+1) \min \left\{ 1, \frac{96 \cdot 10^{27} H_m + 1}{\sqrt{k}} \right\}, \quad (9.2)$$

and

$$\frac{y_k}{x_k} \leq \frac{(192 \cdot 10^{27} H_{8k} + 1)(k+1)}{\sqrt{k}}. \quad (9.3)$$

Proof. Reordering the spectrum does not change either quantity, so assume the order of Section 1. Positivity and Lemma 8.2 permit division by x_k and F .

First suppose $s \geq \sqrt{k}F/4$. The diagonal lower bound gives

$$\frac{y_k}{x_k} \leq \frac{(k+1)F}{s} \leq \frac{4(k+1)}{\sqrt{k}}. \quad (9.4)$$

Otherwise let q count the first k eigenvalues strictly smaller than $\sqrt{k}F$. Each such index contributes more than $1/(1 + \sqrt{k})$ to the sum in (8.1); consequently

$$q \leq (1 + \sqrt{k}) \frac{s}{F} < \frac{k + \sqrt{k}}{4} \leq \frac{k}{2}.$$

Thus $0 \leq q \leq \lfloor k/2 \rfloor$ and $a_q = \lambda_{k-q} \geq \sqrt{k}F$. Put $u = s/F \geq 1$. Also $q \leq 2\sqrt{k}u$, so Proposition 6.2 gives

$$\frac{U_q}{F} = \frac{ku}{C_1(q + \sqrt{k})H_m} \geq \frac{\sqrt{k}u}{C_1(2u+1)H_m} \geq \frac{\sqrt{k}}{3C_1H_m}.$$

The function $(a, z) \mapsto az/(a+z)$ increases in each positive argument. Combining this with $a_q \geq \sqrt{k}F$ and (6.3) gives

$$x_k \geq \frac{\sqrt{k}F}{3C_1H_m + 1}. \quad (9.5)$$

Since $3C_1H_m + 1 = 96 \cdot 10^{27}H_m + 1 > 4$, inequalities (9.4)–(9.5), together with Lemma 8.2, prove (9.2).

For the long-tail version, first use the diagonal bound if $s \geq \sqrt{k}F/8$, obtaining $y_k/x_k \leq 8(k+1)/\sqrt{k}$. In the remaining case choose q by the same threshold. Now

$$q \leq (1 + \sqrt{k}) \frac{s}{F} < \frac{k + \sqrt{k}}{8} \leq \frac{k}{4}, \quad a_q \geq \sqrt{k}F.$$

Thus Proposition 7.1 applies. Again $u = s/F \geq 1$ and $q \leq 2\sqrt{k}u$, giving

$$\frac{V_q}{F} = \frac{ku}{2C_1(q + \sqrt{k})H_{8k}} \geq \frac{\sqrt{k}}{6C_1H_{8k}}, \quad x_k \geq \frac{\sqrt{k}F}{6C_1H_{8k} + 1}.$$

Because $6C_1H_{8k} + 1 = 192 \cdot 10^{27}H_{8k} + 1 > 8$, these two cases prove (9.3). No lower bound on k was imposed in this argument. If one of the small-tail cases is impossible for a small value of k , the corresponding diagonal case already applies; $q = 0$ remains allowed in every construction.

Finally,

$$H_{8k} - H_k \leq \int_k^{8k} \frac{dt}{t} = \log 8 < 3, \quad H_{8k} \leq H_k + 3 \leq 4H_k.$$

If $m \leq k$, use (9.2), whose coefficient $96 \cdot 10^{27} H_m + 1$ is at most $768 \cdot 10^{27} H_m + 1$. If $m \geq k$, use (9.3) and $H_{8k} \leq 4H_k$. Combining either bound with the elementary factor $k + 1$ proves (9.1). This completes the proof. $\square$

Since $H_d \leq 1 + \log d$, the bound is of order $\sqrt{k} [1 + \log \min(k, n - k)]$, with the displayed explicit constant. It improves the elementary factor $k + 1$ only when $(768 \cdot 10^{27} H_{r_*} + 1)/\sqrt{k} < 1$; the constant must be retained when assessing quantitative usefulness. There is no restriction on the condition number, the number of spectral levels, or their multiplicities, and no limiting-noise argument.

The associated dimension-independent question concerns

$$C = \sup_{\substack{n \geq 2, 1 \leq k < n \\ \lambda \in (0, \infty)^n}} \frac{y_k(\lambda)}{x_k(\lambda)}.$$

The theorem does not determine whether C is finite or infinite, and gives no matching lower growth. When $m < k$, the prefactor $(k + 1)/\sqrt{k} = (1 + 1/k)\sqrt{k/m}\sqrt{m}$ also records the remaining imbalance relative to a bound expressed only through the smaller dimension. No symmetry exchanging k and m for arbitrary spectra has been used.

A. Balanced frames and two-level spectra

The frame lemma used in the main proof is designed to tolerate several operations at once. If only square row bases are required, its numerical constant can be substantially improved. For an $n \times d$ Parseval frame Q , define

$$h(Q) = \min_{\substack{I \subseteq [n], |I|=d \\ \det Q_I \neq 0}} \|Q_I^{-1}\|_F^2, \quad M_{n,d} = \sup_{Q^\top Q = I_d} h(Q).$$

There is at least one admissible row basis because Q has column rank d . Singular row subsets are omitted in h ; they will be dealt with explicitly when the frame is used to construct a positive covariance.

Theorem A.1 (A square-basis lower bound). *For every integer $k \geq 4$, there is a real $2k \times k$ Parseval frame Q for which every k -row submatrix is invertible and*

$$h(Q) \geq \frac{k^{3/2}}{4 \cdot 10^9}.$$

Consequently $M_{2k,k} \geq k^{3/2}/(4 \cdot 10^9)$.

Proof. Let Z be a $2k \times k$ standard Gaussian matrix. The $q = 0$ case of Lemma 3.1 gives, for each fixed k -row subset I ,

$$\mathbb{P} \left\{ \|Z_I^{-1}\|_F^2 \leq \frac{\sqrt{k}}{1000} \right\} \leq 2e^{-31k}.$$

A union bound over at most 4^k subsets bounds the probability of any such failure by $2e^{-29k}$. Every one of these square submatrices is invertible almost surely by the polynomial argument in Section 2.

It remains to control the normalization. A $1/2$ -net and the Gaussian moment formula give

$$\mathbb{P}\{\|Z\|_2 > 10\sqrt{k}\} \leq 5^k 2^k e^{-25k/4} < e^{-3k}.$$

Use next a $1/20000$ -net with at most 40001^k points. At any fixed net point v , the negative exponential moment with parameter 10^6 gives

$$\mathbb{P}\{\|Zv\|^2 \leq k/10^6\} \leq e^k(1 + 2 \cdot 10^6)^{-k} \leq (e/(2 \cdot 10^6))^k.$$

The probability of a failure somewhere on the net is therefore at most $(40001e/(2 \cdot 10^6))^k < 16^{-k}$; the last inequality follows from $e < 3$ and $16 \cdot 40001 \cdot 3 < 2 \cdot 10^6$. Off the two net exceptional sets, the triangle inequality gives, for every unit vector x ,

$$\|Zx\| \geq \frac{\sqrt{k}}{1000} - \frac{10\sqrt{k}}{20000} = \frac{\sqrt{k}}{2000}.$$

Thus $Z^\top Z \geq kI/(4 \cdot 10^6)$. The total exceptional probability is at most $e^{-3k} + 16^{-k} + 2e^{-29k} < 1$ for $k \geq 4$. Choose a realization satisfying all the required inequalities and ranks, and put $Q = Z(Z^\top Z)^{-1/2}$. Then $Q^\top Q = I_k$, and for every I ,

$$\begin{aligned} \|Q_I^{-1}\|_F^2 &= \text{tr}(Z_I^{-\top}(Z^\top Z)Z_I^{-1}) \\ &\geq \frac{k}{4 \cdot 10^6} \|Z_I^{-1}\|_F^2 \geq \frac{k^{3/2}}{4 \cdot 10^9}. \end{aligned}$$

This proves the assertion by a positive-probability existence argument. $\square$

Proposition A.2 (Replication and complementary rank). *Put $c_0 = (16 \cdot 10^9)^{-1}$. For $n \geq 2d \geq 2$,*

$$M_{n,d} \geq c_0 n \sqrt{d}.$$

For $m = n - k$ and $r = \min(k, m)$, one consequently has

$$m - k + M_{n,k} \geq \max(m - k, 0) + c_0 n \sqrt{r}.$$

In both assertions the lower bound is realized by an actual frame.

Proof. For $d \geq 4$, take a frame supplied by Theorem A.1. Replace each row by $u = \lfloor n/(2d) \rfloor$ copies divided by $\sqrt{u}$, and append zero rows to reach n rows. An invertible row basis contains no zero row and no two copies of an original row. Its inverse squared Frobenius norm is therefore u times the original one. Since $u \geq n/(4d)$, the first assertion follows. For $d = 1, 2, 3$, partition the n rows into d nonempty groups of sizes $n_1, \dots, n_d$. In group j use n_j copies of $e_j^\top / \sqrt{n_j}$. The frame is Parseval, and every invertible basis has inverse squared Frobenius norm $\sum_j n_j = n$, which suffices.

For completeness, complementary frames satisfy the exact identity

$$M_{n,k} = M_{n,m} + k - m. \quad (\text{A.1})$$

Indeed, complete a Parseval frame Q to an orthogonal matrix $[Q \ R]$. For complementary row sets I, J of cardinalities k, m ,

$$Q_I Q_I^\top = I_k - R_I R_I^\top, \quad R_J^\top R_J = I_m - R_I^\top R_I.$$

If $k \geq m$, a singular value decomposition of R_I shows that these matrices are invertible together and that the first has the eigenvalues of the second together with $k - m$ additional ones. Their inverse traces therefore differ by $k - m$. Interchanging the roles of the frames proves the same statement when $k < m$. Taking minima over complementary bases and then suprema proves (A.1). Apply the first assertion to the smaller column dimension. Complementation preserves the existence of a frame attaining the stated lower bound. $\square$

Theorem A.3 (Two-level spectral consequences). *Let $m = n - k$, $r = \min(k, m)$, and $c_0 = (16 \cdot 10^9)^{-1}$. For each fixed n, k ,*

$$\limsup_{\varepsilon \downarrow 0} \frac{y_k(1^k, \varepsilon^m)}{x_k(1^k, \varepsilon^m)} \leq \frac{(k+1)m}{\max(m-k, 0) + c_0 n \sqrt{r}} \leq 16 \cdot 10^9 \left(\sqrt{r} + \frac{1}{\sqrt{r}} \right). \quad (\text{A.2})$$

Here an exponent in the spectral list denotes repetition. In balanced dimensions, for every $k \geq 4$ and every $0 < \varepsilon \leq 1$,

$$x_k(1^k, \varepsilon^k) \leq y_k(1^k, \varepsilon^k) \leq \frac{(4 \cdot 10^9 + 1)(k+1)}{\sqrt{k}} x_k(1^k, \varepsilon^k). \quad (\text{A.3})$$

Proof. For a rank- k projection $P = QQ^\top$ set $K_\varepsilon = \varepsilon I_n + (1 - \varepsilon)P$. Its spectrum is $(1^k, \varepsilon^m)$, and direct multiplication gives $K_\varepsilon^2 = (1 + \varepsilon)K_\varepsilon - \varepsilon I_n$. Cyclicity of trace in the Nyström formula therefore yields, for every k -subset I ,

$$E_{K_\varepsilon}(I) = \varepsilon \left[m - k + \text{tr}((K_\varepsilon)_{I,I}^{-1}) \right]. \quad (\text{A.4})$$

For invertible Q_I , the bracket converges to $m - k + \|Q_I^{-1}\|_F^2$. If Q_I is singular, the inverse trace diverges because $(K_\varepsilon)_{I,I}$ has an eigenvalue ε . There are finitely many subsets and at least one invertible row basis, so their minimum converges to $m - k + h(Q)$. Choose the actual frame from Proposition A.2. Maximizing the error over orientations gives

$$\liminf_{\varepsilon \downarrow 0} \frac{x_k(1^k, \varepsilon^m)}{\varepsilon} \geq \max(m - k, 0) + c_0 n \sqrt{r}.$$

For fixed n, k , elementary product counting gives $e_k(1^k, \varepsilon^m) = 1 + O(\varepsilon)$ and $e_{k+1}(1^k, \varepsilon^m) = m\varepsilon + O(\varepsilon^2)$. Thus $y_k/\varepsilon \rightarrow (k+1)m$, proving the first inequality in (A.2). If $k \leq m$, then $(k+1)m/(n\sqrt{k}) \leq (k+1)/\sqrt{k}$. If $k \geq m$, then $k+1 \leq n$ and $(k+1)m/(n\sqrt{m}) \leq \sqrt{m}$. Dropping the nonnegative term $\max(m-k, 0)$ from the denominator proves the second inequality.

We now keep ε positive and assume $m = k \geq 4$. Let Q be from Theorem A.1, and put $L = k^{3/2}/(4 \cdot 10^9)$. More generally the following argument applies to any frame with $h(Q) \geq L$. For an invertible basis let t_i be the inverse eigenvalues of $Q_I Q_I^\top$ and $T = \sum_i t_i \geq L$. Equation (A.4) gives

$$\frac{E_{K_\varepsilon}(I)}{\varepsilon} = \sum_i \frac{t_i}{1 - \varepsilon + \varepsilon t_i} \geq \frac{T}{1 + \varepsilon T} \geq \frac{L}{1 + \varepsilon L}.$$

If a basis is singular, its error is at least one, which also exceeds $\varepsilon L/(1 + \varepsilon L)$. Together with the diagonal orientation this yields

$$x_k(1^k, \varepsilon^k) \geq \max \left\{ k\varepsilon, \frac{\varepsilon L}{1 + \varepsilon L} \right\}. \quad (\text{A.5})$$

Two upper bounds on the symmetric quotient suffice:

$$y_k(1^k, \varepsilon^k) \leq \min\{k(k+1)\varepsilon, (k+1)\sqrt{\varepsilon}\}. \quad (\text{A.6})$$

For the first, count products to obtain

$$e_k = \sum_{j=0}^k \binom{k}{j}^2 \varepsilon^j, \quad e_{k+1} = \varepsilon \sum_{j=0}^{k-1} \binom{k}{j} \binom{k}{j+1} \varepsilon^j;$$

then use $\binom{k}{j+1} \leq k \binom{k}{j}$. For the second put $b = (1 + \varepsilon)/\sqrt{\varepsilon} \geq 2$. After rescaling the variable in $(1 + t)^k(1 + \varepsilon t)^k$, the polynomial becomes

$$(1 + bu + u^2)^k = \sum_{j=0}^k \binom{k}{j} (b-2)^j u^j (1+u)^{2(k-j)}.$$

In each summand the coefficient of u^{k+1} is no greater than that of u^k . Since all weights are non-negative, the same is true for the sum. The coefficient of u^a is $e_a \varepsilon^{-a/2}$, proving the second bound in (A.6).

If $\varepsilon \leq k^{-2}$, use the first quotient bound and the second term of (A.5) to get

$$\frac{y_k}{x_k} \leq (k+1)(k/L + k\varepsilon) \leq (k+1)(k/L + 1/\sqrt{k}).$$

For $k^{-2} \leq \varepsilon \leq k^{-1}$ use the second quotient bound to get

$$\frac{y_k}{x_k} \leq (k+1) \left(\frac{1}{L\sqrt{\varepsilon}} + \sqrt{\varepsilon} \right) \leq (k+1)(k/L + 1/\sqrt{k}).$$

For $\varepsilon \geq k^{-1}$ the diagonal lower bound gives $y_k/x_k \leq (k+1)/(k\sqrt{\varepsilon}) \leq (k+1)/\sqrt{k}$. Substitute the value of L . The lower comparison in (A.3) is Lemma 8.2. $\square$

These are upper bounds on the comparison ratio for two-level spectra. The lower bound on $M_{2k,k}$ rules out a uniform $o(k^{3/2})$ upper bound on that frame extremum, but it does not establish divergence of the unrestricted spectral comparison ratio.

B. A quantitative shifted-rank envelope

A version of the projection argument with fewer simultaneous requirements improves its constant in balanced ambient dimensions. It also retains a sharper denominator than scalar regularization with the coarser envelope used in the main proof.

Proposition B.1 (Shifted-rank projection and spectral bound). *Let $k \geq 4$, $0 \leq q \leq \lfloor k/2 \rfloor$, and*

$$L = \frac{k^2}{10^{23}(q + \sqrt{k})}.$$

There is a rank- $(k+q)$ orthogonal projection P on $\mathbb{R}^{2k}$ for which every k -principal block is positive definite and has inverse trace at least L . For any ordered positive spectrum of length $2k$, put $a = \lambda_{k-q}$, $b = \lambda_{2k}$ and $s = \sum_{i=k+1}^{2k} \lambda_i$. Then

$$x_k(\lambda) \geq \max \left\{ s, \frac{abL}{a - b + bL} \right\}. \quad (\text{B.1})$$

If $a_1, \dots, a_{k-q}$ and $b_1, \dots, b_{k+q}$ are fixed positive numbers, $b_ = \min_j b_j$, $b^* = \max_j b_j$, and $\theta = b^*/b_*$, then*

$$\begin{aligned} \limsup_{\varepsilon \downarrow 0} \frac{y_k(a_1, \dots, a_{k-q}, \varepsilon b_1, \dots, \varepsilon b_{k+q})}{x_k(a_1, \dots, a_{k-q}, \varepsilon b_1, \dots, \varepsilon b_{k+q})} &\leq \frac{(k+1)e_{q+1}(b)}{b_* L e_q(b)} \\ &\leq \frac{10^{23}\theta(k+1)}{\sqrt{k}}. \end{aligned} \quad (\text{B.2})$$

Proof. We first sharpen the Gaussian threshold for the present dimensions. For a $k \times (k + q)$ standard Gaussian matrix G , the proof of Lemma 3.1, now taking $j = \lfloor \sqrt{k} \rfloor$, gives

$$\mathbb{P} \left\{ \text{tr}((GG^\top)^{-1}) \leq \frac{k}{1000(q + \sqrt{k})} \right\} \leq 2e^{-31k}. \quad (\text{B.3})$$

Indeed, a trace below the displayed threshold forces at least half the j -subsets of diagonal entries to have sum at most $j/[500(q + \sqrt{k})]$. Conditioning on the other rows identifies each such block as the inverse Gram matrix of a $j \times (q + j)$ standard Gaussian matrix W . Cauchy–Schwarz then forces $\|W\|_F^2 \geq 500j(q + \sqrt{k}) \geq 500j(q + j)$. The exponential-moment bound is $e^{-124j(q+j)} \leq e^{-31k}$ because $j^2 \geq k/4$. Row exchangeability supplies the factor two.

Let Z be a $2k \times p$ standard Gaussian matrix, $p = k + q \leq 3k/2$. The $1/2$ -net argument gives

$$\mathbb{P}\{\|Z\|_2 > 10\sqrt{k}\} \leq 5^p 2^k e^{-25k/4} < e^{-2k}.$$

Set $D_* = 10^{10}$ and use a $1/(20D_*)$ -net of at most $(41D_*)^{3k/2}$ points. At any fixed point the probability of $\|Zv\|^2 \leq 4k/D_*^2$ is at most $(2e/D_*^2)^k$. The union bound is

$$\left(\frac{2e 41^{3/2}}{\sqrt{D_*}} \right)^k < (1722/100000)^k < 16^{-k}.$$

Off these events the triangle inequality gives $Z^\top Z \geq kI/D_*^2$. Equation (B.3) and at most 4^k row subsets add failure probability at most $2e^{-29k}$. The sum is less than one. Choose a realization with all these inequalities and all relevant row ranks. Set $V = Z(Z^\top Z)^{-1/2}$ and $P = VV^\top$. Every k -principal block is positive definite, and

$$P_{J,J} \leq (D_*^2/k)Z_J Z_J^\top, \quad \text{tr}(P_{J,J}^{-1}) \geq \frac{k^2}{1000D_*^2(q + \sqrt{k})} = L.$$

Assign the smallest $k + q$ covariance eigenvalues to the range of P and the largest $k - q$ to its orthogonal complement. The resulting precision satisfies

$$K^{-1} \leq a^{-1}(I - P) + b^{-1}P = a^{-1}I + (b^{-1} - a^{-1})P.$$

For a principal block, with inverse eigenvalues t_i of $P_{J,J}$ and $T = \sum_i t_i \geq L$, inverse order yields

$$\text{tr}(((K^{-1})_{J,J})^{-1}) \geq \sum_i \frac{t_i}{b^{-1} - a^{-1} + a^{-1}t_i} \geq \frac{L}{b^{-1} - a^{-1} + a^{-1}L}.$$

The argument also applies when $a = b$: then the denominator is a positive multiple of t_i , and the last scalar function is constant. The precision identity and the diagonal orientation prove (B.1).

Finally, for small ε , the unscaled values are the leading $k - q$ eigenvalues and the smallest eigenvalue is εb_* . The lower bound implies $\liminf x_k/\varepsilon \geq b_* L$. Product expansions give

$$e_k = (\prod_i a_i) \varepsilon^q e_q(b) + O(\varepsilon^{q+1}), \quad e_{k+1} = (\prod_i a_i) \varepsilon^{q+1} e_{q+1}(b) + O(\varepsilon^{q+2}).$$

Dividing the resulting limit of y_k/ε by the positive lower limit for x_k/ε proves the first inequality in (B.2). Counting extensions of a q -subset gives $(q + 1)e_{q+1}(b) \leq kb_* e_q(b)$. The resulting bound is $10^{23}\theta(k+1)(q + \sqrt{k})/[k(q+1)]$, which is at most the stated quantity because $q + \sqrt{k} \leq (q+1)\sqrt{k}$. $\square$

C. Spectral capping and a finite-level reduction

The comparison problem can be restricted to spectra with controlled range at a fixed loss. The reduction is independent of the Gaussian construction and does not require balanced dimensions.

Theorem C.1 (Capping, deletion, and dyadic rounding). *Let $\lambda \in (0, \infty)^n$, $1 \leq k < n$, and put*

$$F = \frac{e_{k+1}(\lambda)}{e_k(\lambda)}, \quad T = (k+1)F, \quad \lambda_i^c = \min(\lambda_i, T).$$

Then

$$\begin{aligned} y_k(\lambda^c) &\geq \frac{1}{2}y_k(\lambda), & x_k(\lambda^c) &\leq x_k(\lambda), & \max_i \lambda_i^c &\leq 2y_k(\lambda^c), \\ \frac{y_k(\lambda)}{x_k(\lambda)} &\leq 2 \frac{y_k(\lambda^c)}{x_k(\lambda^c)}. \end{aligned} \tag{C.1}$$

Let $\delta = F/(4n)$, and retain only the entries of λ^c that are at least δ , obtaining a vector μ of length d . Then

$$k < d \leq n, \quad \delta \leq \mu_i \leq T, \quad y_k(\mu) \geq \frac{1}{4}y_k(\lambda), \quad x_k(\mu) \leq x_k(\lambda). \tag{C.2}$$

Finally define

$$v_i = \delta 2^{\lfloor \log_2(\mu_i/\delta) \rfloor}.$$

The vector v has at most $1 + \lfloor \log_2(4n(k+1)) \rfloor$ distinct positive entries and satisfies

$$\frac{y_k(\lambda)}{x_k(\lambda)} \leq 8 \frac{y_k(v)}{x_k(v)}. \tag{C.3}$$

The original dimension n , rather than the retained dimension d , occurs in the level bound.

Proof. We first record two monotonicity facts. For a positive list with one entry z distinguished, write $A = e_k$ (remaining list), $B = e_{k+1}$ (remaining list), and $D_0 = e_{k-1}$ (remaining list). Provided the whole list has length at least $k+1$, one has $A, D_0 > 0$, $B \geq 0$, and $A^2 \geq BD_0$ by the coefficient argument in Lemma 8.1. Thus

$$f(z) := \frac{e_{k+1}}{e_k} = \frac{zA + B}{zD_0 + A}, \quad f'(z) = \frac{A^2 - BD_0}{(zD_0 + A)^2} \geq 0. \tag{C.4}$$

Consequently e_{k+1}/e_k and y_k are coordinatewise nondecreasing. They are homogeneous of degree one.

Likewise, for positive spectra of the same length, $v_i \leq \lambda_i$ for every i implies $x_k(v) \leq x_k(\lambda)$. Use the same eigenbasis to obtain $K_v \leq K_\lambda$. Inversion reverses the order, passage to a principal block preserves it, and another inversion reverses it back. The precision identity (2.2) therefore compares the residual errors for every selected set in every orientation. Taking the minimum over subsets and maximum over orientations proves the claim.

Since $f(z) \leq A/D_0$, for any $T > 0$ we also have

$$\frac{1}{f(T)} = \frac{TD_0 + A}{TA + B} \leq \frac{D_0}{A} + \frac{1}{T} \leq \frac{1}{f(z)} + \frac{1}{T}. \tag{C.5}$$

There are at most k entries of λ strictly larger than the chosen $T = (k+1)F$. Otherwise fix $k+1$ such entries. Every k -subset omits at least one of them, and the identity

$$(k+1)e_{k+1}(\lambda) = \sum_{|I|=k} \left(\prod_{i \in I} \lambda_i \right) \sum_{j \notin I} \lambda_j$$

would make the right side strictly larger than $Te_k(\lambda)$, a contradiction. Cap these entries successively at the original value T . Equation (C.5) accumulates to

$$\frac{1}{F_k(\lambda^c)} \leq \frac{1}{F} + \frac{k}{T} = \frac{2k+1}{(k+1)F} < \frac{2}{F},$$

where $F_k = e_{k+1}/e_k$. Thus $F_k(\lambda^c) \geq F/2$. The cap is at most $T \leq 2y_k(\lambda^c)$, and the monotonicity of x_k proves the remaining claims in (C.1).

Let s_{out} be the sum of the entries removed below $\delta = F/(4n)$. Then $s_{\text{out}} < F/4$. At least $k+1$ entries survive: if at most k did, every product in $e_{k+1}(\lambda^c)$ would contain a removed entry, and hence

$$e_{k+1}(\lambda^c) \leq \sum_{i \text{ removed}} \lambda_i^c e_k(\lambda^c \text{ with } i \text{ omitted}) \leq s_{\text{out}} e_k(\lambda^c).$$

This would contradict $F_k(\lambda^c) \geq F/2$.

Deleting an entry z while retaining at least $k+1$ entries loses at most z in the quotient, because

$$f(z) = \frac{zA+B}{zD_0+A} \leq z + \frac{B}{A} = z + F_k(\text{remaining list}).$$

Successive deletions therefore yield $F_k(\mu) \geq F_k(\lambda^c) - s_{\text{out}} > F/4$. This proves the y_k assertion in (C.2), and the bounds on retained entries are immediate.

To compare x_k across dimensions, let $H \geq \delta I_d$ be a positive definite $d \times d$ matrix, $d > k$, and let $z_1, \dots, z_\ell < \delta$ be positive. For $K = \text{diag}(H, z_1, \dots, z_\ell)$ we claim that an optimal k -subset selects only coordinates in H , so

$$\min_{|I|=k} E_K(I) = \sum_{j=1}^{\ell} z_j + \min_{\substack{|I|=k \\ I \subseteq [d]}} E_H(I). \quad (\text{C.6})$$

To prove it, suppose a selected subset contains a scalar coordinate z_j . Remove it, increasing the error by z_j . If S is its remaining selected set inside H , the unselected Schur complement Z of H is at least δI : this follows from $H^{-1} \leq \delta^{-1}I$ and the precision identity. Selecting any still-unselected coordinate i of H decreases its residual trace by

$$\frac{\|Z_{:,i}\|_2^2}{Z_{ii}} \geq Z_{ii} \geq \delta,$$

as follows from the one-coordinate Schur update. Such a coordinate exists since $d > k$. The replacement strictly decreases the total error. Repeating proves (C.6).

Choose H to be an orientation of the retained spectrum μ attaining $x_k(\mu)$, and put the removed entries in scalar blocks. The resulting matrix has spectrum λ^c , so

$$x_k(\lambda) \geq x_k(\lambda^c) \geq s_{\text{out}} + x_k(\mu) \geq x_k(\mu).$$

This completes (C.2). In particular the condition number of μ is at most $T/\delta = 4n(k+1)$.

The dyadic definition gives $\mu_i/2 \leq v_i \leq \mu_i$. Monotonicity and homogeneity give $y_k(v) \geq y_k(\mu)/2 \geq y_k(\lambda)/8$, while $x_k(v) \leq x_k(\mu) \leq x_k(\lambda)$. These prove (C.3). The exponents in the dyadic definition are integers between zero and $\lfloor \log_2(4n(k+1)) \rfloor$, which proves the level count. $\square$

Corollary C.2 (A restricted supremum). *Let C be the supremum of $y_k(\lambda)/x_k(\lambda)$ over all positive spectra and allowed dimensions, and let C_{cap} restrict the same supremum to spectra satisfying $\max_i \lambda_i \leq 2y_k(\lambda)$. Then, as extended-real inequalities,*

$$C_{\text{cap}} \leq C \leq 2C_{\text{cap}}.$$

In particular C is finite if and only if C_{cap} is finite.

Proof. The first inequality is inclusion of the domains of the suprema. For each spectrum, its capped spectrum belongs to the restricted domain, and (C.1) bounds its original ratio by twice that restricted ratio. Take the supremum. $\square$

The reduction preserves the finite-versus-infinite question, but does not answer it. Its logarithmic level count depends on the original dimension and is not a reduction to a fixed number of levels, in particular not to two-level spectra.

D. A dimensional comparison and its strict improvement

The estimate in this appendix has a smaller explicit constant in finite dimensions than Theorem 9.1. It is independent of the Gaussian construction. Water-level functions and symmetric-quotient inequalities also occur in A-optimal principal-submatrix selection; see [9, Definition 4 and Lemma 9]. Those scalar ingredients are background. Here they are combined with a hard orientation and a quantitative analysis of near equality to improve a uniform orbit comparison strictly for every fixed interior pair of dimensions.

Proposition D.1 (Harmonic blocks). *Order $\lambda_1 \geq \dots \geq \lambda_n > 0$ and put $m = n - k$. For $k + 1 \leq q \leq n$, define*

$$H_q = \frac{q}{\sum_{i \leq q} \lambda_i^{-1}}, \quad B_q = \sum_{i > q} \lambda_i + (q - k)H_q.$$

Then $x_k \geq \max_q B_q$. There is a unique $\tau > 0$ satisfying

$$\sum_{i=1}^n (1 - \tau/\lambda_i)_+ = k.$$

Set $q = \#\{i : \lambda_i > \tau\}$, $d = q - k$, and $T = \sum_{i > q} \lambda_i$. Then $q \geq k + 1$, $d \geq 1$, and

$$B := \max_j B_j = q\tau + T, \quad y_k \leq (k + 1) \frac{d\tau + T}{q\tau + T} x_k \leq \frac{(k + 1)m}{n} x_k. \quad (\text{D.1})$$

The last inequality is strict at every positive spectrum if $m \geq 2$. For $k = n - 1$ one has $x_k = y_k = n/(\sum_i \lambda_i^{-1})$.

Proof. The zero-diagonal induction used in Section 7 constructs a q -dimensional positive covariance block with the first q eigenvalues and constant precision diagonal H_q^{-1} . Append the other eigenvalues as singleton blocks. Leaving r coordinates in the first block incurs inverse trace at least rH_q , by $\text{tr}(A^{-1}) \text{tr}(A) \geq r^2$. Since $H_q \geq \lambda_q$, the smallest resulting lower bound for m unselected coordinates leaves every tail singleton and $q - k$ coordinates in the block. This proves $x_k \geq B_q$.

The water-level equation is continuous and strictly decreasing from n to zero on $(0, \lambda_1)$. Its q nonzero summands are each less than one, whence $q > k$; it gives $\sum_{i \leq q} \tau/\lambda_i = d$, $T \leq (n - q)\tau$, and $B_q = q\tau + T$. To check maximality, put $S_j = \sum_{i \leq j} \lambda_i^{-1}$ and $z = \lambda_{j+1}S_j$. Direct subtraction gives

$$\frac{B_{j+1} - B_j}{\lambda_{j+1}} = -\frac{(z - j)(z - (j - k))}{z(z + 1)}.$$

Here $z \leq j$, and the threshold $\lambda_{j+1}S_j - (j - k)$ is nonincreasing in j , because $\lambda_{j+2}S_{j+1} - (j + 1 - k) \leq \lambda_{j+1}S_j - (j - k)$. The water-level index is where that threshold crosses zero; hence it maximizes B_j , allowing ties.

Put $\beta_i = \tau/\lambda_i \in (0, 1)$ for $i \leq q$. Since $\sum_i \beta_i = d$, counting products gives

$$de_d(\beta) = \sum_{|S|=d-1} \beta_S \sum_{j \notin S} \beta_j \geq e_{d-1}(\beta).$$

The inequality is strict if $d \geq 2$. Complementation of symmetric polynomials bounds the quotient of the first q entries by $d\tau$. Adding a positive entry a increases the quotient by strictly less than a , since

$$\frac{e_{k+1} + ae_k}{e_k + ae_{k-1}} < \frac{e_{k+1}}{e_k} + a.$$

Consequently $F := e_{k+1}(\lambda)/e_k(\lambda) \leq d\tau + T$, strictly if $d \geq 2$ or $n - q > 0$. Finally

$$\frac{m}{n}(q\tau + T) - (d\tau + T) = \frac{k}{n}((n - q)\tau - T) \geq 0.$$

This proves the assertions, including strictness when $m \geq 2$ and coincident bounds when $m = 1$. $\square$

The constant in (D.1) is sharp for y_k/B : for $(1^k, \varepsilon^m)$, the water level is $m\varepsilon/(m + k\varepsilon)$, so $B \sim n\varepsilon$ while $y_k \sim (k + 1)m\varepsilon$. Sharpness for this auxiliary lower bound does not imply sharpness for y_k/x_k . The following theorem separates them uniformly.

Theorem D.2 (An explicit uniform strict improvement). *Let $k, m \geq 2$, $n = k + m$, and define*

$$H = \frac{3k + m - 2 + \sqrt{(n - 2)^2 + 4km}}{2}, \quad g = H - 2k, \quad N = n + g, \\ L = \max\{2, N, 4H^2/g\},$$

$$\eta = \min\left\{\frac{k}{4n}, \frac{1}{2^{n+1}L}, \frac{kg}{4nN}, \frac{mg}{2n + g}\right\}.$$

Then $g, \eta > 0$, and every positive spectrum satisfies

$$y_k \leq \frac{(k + 1)(m - \eta)}{n} x_k < \frac{(k + 1)m}{n} x_k. \quad (\text{D.2})$$

Proof. Normalize the water level to one, using homogeneity. Adopt q, d, T, B from Proposition D.1, and set $t = n - q = m - d$, $\beta_i = \lambda_i^{-1} \in (0, 1)$ for $i \leq q$, and $a_j = \lambda_{q+j} \in (0, 1]$. Thus $\sum \beta_i = d$ and

$$U = t - T = \sum_j (1 - a_j), \quad B = n - U, \quad D = \frac{m}{n}B - F = \frac{k}{n}U + (d + T - F) \geq 0. \quad (\text{D.3})$$

In particular $U \leq nD/k$ and $B \leq n$.

Stability. Put $V = \sum_i \beta_i(1 - \beta_i)$. We claim $V \leq 2^n D$ whenever $D < k/(2n)$. If $d \geq 2$, expansion gives

$$de_d(\beta) - e_{d-1}(\beta) = \sum_i \beta_i(1 - \beta_i)e_{d-2}(\beta_{-i}).$$

Every last symmetric polynomial is at least one. Indeed, entries in $[0, 1]$ whose sum is at least an integer r have $e_r \geq 1$: move pairs of fractional entries of fixed sum toward endpoints. This cannot increase e_r , whose dependence on their product has a nonnegative coefficient, and leaves at least r ones. Apply this with $r = d - 2$ and $\sum \beta_{-i} = d - \beta_i \geq d - 1$. Since $e_d(\beta) \leq \binom{q}{d} \leq 2^n$, the active quotient's deficit below d is at least $V/2^n$. Adding tails increases the quotient by at most T , proving the claim.

If $d = 1$, then $t = m - 1 \geq 1$ and $U < 1/2$, so every tail entry exceeds $1/2$. The active quotient is one, and its preceding coefficient ratio is $e_2(\beta) = V/2$. Adding a tail entry $a > 1/2$ loses, relative to the additive upper bound $1 + a$, the amount

$$(1 + a) - \frac{1 + a}{1 + aV/2} = \frac{a(1 + a)V}{2 + aV} \geq \frac{V}{4}.$$

Here $V \leq 1$. Later additions preserve this accumulated deficit; thus $D \geq V/4 \geq V/2^n$.

Round each β_i to its nearest endpoint. The sum of rounding errors is at most $2V \leq 2^{n+1}D$. If this sum is less than one, exactly d entries round to one, because the original sum and the number of rounded ones are integers. Exactly k entries therefore round to zero. If $2^{n+1}D \leq L^{-1} < 1$, these entries are at most L^{-1} , while every tail eigenvalue is at least $1 - nD/k$. Ordering the covariance eigenvalues gives

$$\lambda_i \geq L \ (i \leq k), \quad \lambda_i \geq 1 - nD/k \ (i > k). \quad (\text{D.4})$$

A frame with equal inverse-basis cost. Set $r_0 = m$ and $r_1 = \dots = r_k = 1$. The equation

$$\frac{m}{c + 2m} + \frac{k}{c + 2} = 1$$

has a unique positive solution $c = (k - m - 2 + \sqrt{(n - 2)^2 + 4km})/2$, with $n + c = H$. Put $p_i = (c + 2r_i)^{-1}$, $Z = \sum p_i$, $z_i = p_i/Z$, and $u_i = \sqrt{z_i}$. Choose a $(k + 1) \times k$ Parseval matrix W with $WW^\top = I - uu^\top$. Repeat row i exactly r_i times, dividing each copy by $\sqrt{r_i}$, to obtain an $n \times k$ Parseval matrix Q . Every independent k -row selection uses distinct groups and omits one group i . Sherman–Morrison gives

$$\|Q_I^{-1}\|_F^2 = n - 2r_i + \frac{\sum_j r_j z_j}{z_i} = n + c = H,$$

since $\sum_j r_j p_j = 1$. Further, $(n - 2)^2 + 4km - (k - m + 2)^2 = 8k(m - 1) > 0$, which proves $g > 0$.

For $0 < a \leq 1$, put $P = QQ^\top$ and $K = LP + a(I - P)$. If $G = P_{I,I}$ is invertible, its eigenvalues $\gamma_j \in (0, 1]$ have $\sum \gamma_j^{-1} = H$. With $\epsilon = a/L$, the exact two-level residual identity yields

$$\begin{aligned} E_K(I) &= a \left[m - k + \sum_j (\gamma_j + \epsilon(1 - \gamma_j))^{-1} \right] \\ &\geq a(m - k + H) - a\epsilon H^2 \geq aN - H^2/L. \end{aligned}$$

We used $(1 + s)^{-1} \geq 1 - s$ and $\sum \gamma_j^{-2} \leq H^2$. If G is singular, the same formula instead gives $E_K(I) \geq L + a(m - 1) \geq N$, also sufficient. Hence

$$x_k(L^k, a^m) \geq aN - H^2/L. \quad (\text{D.5})$$

Dichotomy. If $D \geq \eta$, then $x_k \geq B$ and (D.3) imply $y_k/x_k \leq (k + 1)(m/n - D/B) \leq (k + 1)(m - \eta)/n$. Otherwise the first three constraints defining η imply (D.4) with $a = 1 - g/(4N) \in (0, 1)$. Coordinatewise spectral monotonicity of x_k , proved in Appendix C, and (D.5) give

$$x_k(\lambda) \geq aN - H^2/L \geq n + g/2.$$

Meanwhile $F \leq (m/n)B \leq m$. The last constraint $\eta \leq mg/(2n + g)$ gives

$$\frac{y_k}{x_k} \leq \frac{(k + 1)m}{n + g/2} \leq \frac{(k + 1)(m - \eta)}{n}.$$

This completes the proof. □

The same stability proof classifies near equality in the harmonic comparison. For fixed n, k with $k, m \geq 2$, if $y_k/B \rightarrow (k+1)m/n$ and the water level is normalized to one, then $D \rightarrow 0$, exactly k eigenvalues tend to infinity, and every remaining eigenvalue tends to one. Their divergence rates need not coincide. The finite bound (D.5) covers all those rates. The positive improvement η is conservative and includes an exponential factor; no optimality is asserted.

E. Arbitrary weighted tails and anisotropic rank boundaries

This appendix retains estimates that are stronger near a rank boundary than the unrestricted logarithmic comparison. The boundary coefficients are fixed before the regularization tends to zero; they may have any condition number. A finite positive spectrum requires the displayed tail-trace or leading-condition-number hypothesis for the corresponding stronger conclusion.

Theorem E.1 (Weighted complementary frames). *Put $c_0 = (16 \cdot 10^9)^{-1}$, $m = n - k$, $r = \min(k, m)$, and*

$$\gamma(k, m) = \begin{cases} c_0 k / (2\sqrt{m}), & m \leq k, \\ c_0 \sqrt{k} / 8, & m > k. \end{cases}$$

For every positive diagonal $m \times m$ matrix B there is a real $n \times m$ Parseval matrix R such that every nonsingular row basis satisfies

$$\text{tr}(B(R_J^\top R_J)^{-1}) \geq \gamma(k, m) \text{tr} B. \quad (\text{E.1})$$

For any ordered positive spectrum, put $a = \lambda_k$ and $s = \sum_{i>k} \lambda_i$. Then

$$x_k \geq \max \left\{ s, \frac{a\gamma(k, m)s}{a + \gamma(k, m)s} \right\}, \quad \frac{y_k}{x_k} \leq (k+1) \min\{1, \gamma(k, m)^{-1} + s/a\}. \quad (\text{E.2})$$

In particular, $s/a \leq \sqrt{r}/(k+1)$ implies $y_k/x_k \leq (256 \cdot 10^9 + 1)\sqrt{r}$. For fixed positive lists $a_1, \dots, a_k$ and $b_1, \dots, b_m$,

$$\limsup_{\epsilon \downarrow 0} \frac{y_k(a_1, \dots, a_k, \epsilon b_1, \dots, \epsilon b_m)}{x_k(a_1, \dots, a_k, \epsilon b_1, \dots, \epsilon b_m)} \leq \frac{k+1}{\gamma(k, m)} \leq 256 \cdot 10^9 \sqrt{r}. \quad (\text{E.3})$$

Proof. Proposition A.2 and its complementary identity supply, for every $t \geq 1$, a $(k+t) \times t$ Parseval matrix whose row-basis inverse costs are at least

$$c_0(k+t)\sqrt{\min(k, t)} + \max(t-k, 0). \quad (\text{E.4})$$

Order the diagonal weights $b_1 \geq \dots \geq b_m > 0$, with total s . For each t , put its first t columns in this frame and the remaining columns in singleton identity blocks. An invertible m -row selection must include every singleton; its weighted cost is at least $c_0(k+t)\sqrt{\min(k, t)}b_t$.

If $m \leq k$, let $W = \max_{t \leq m} (k+t)\sqrt{t} b_t$. The inequality $\sum_{t \leq m} t^{-1/2} \leq 2\sqrt{m}$ gives $s \leq 2W\sqrt{m}/k$, proving (E.1). If $m > k$ and the first $h = \min(m, 8k)$ weights have total at least $s/2$, let $W = \max_{t \leq h} (k+t)\sqrt{\min(k, t)}b_t$. Their first k terms total at most $2W/\sqrt{k}$, and the remaining terms total at most

$$\frac{W}{\sqrt{k}} \sum_{t=k+1}^{8k} \frac{1}{k+t} \leq \frac{W}{\sqrt{k}} \log(9/2) < \frac{2W}{\sqrt{k}}.$$

Thus $W \geq s\sqrt{k}/8$, which also proves the claimed bound.

It remains to treat $m > 8k$ with the first $8k$ weights summing to less than $s/2$. If $k \leq 3$, every Parseval row basis has weighted cost at least $s \geq \gamma(k, m)s$, so assume $k \geq 4$. Take the full-spark $2k \times k$ frame Q_0 from Theorem A.1, with inverse costs at least $4c_0k^{3/2}$. After the first $8k$ weights,

the remaining $m' = m - 8k$ weights total $s' > s/2$ and each is less than $s/(16k)$; hence $m' > 8k$. Reserve the k smallest, whose total is at most $ks'/m' < s'/8$. Greedily distribute the others among $2k$ groups of currently smallest total. The totals differ by at most the largest weight, so all groups are nonempty and their totals satisfy

$$v_j \geq \frac{7s}{32k} - \frac{s}{16k} > \frac{s}{8k}. \quad (\text{E.5})$$

For each group the zero-diagonal construction of Section 7 provides a unit vector u_j with nonzero entries and a positive semidefinite matrix D_j , with the group's weights as its positive spectrum, $\ker D_j = \text{span}(u_j)$, and $(D_j)_{ii} = v_j(u_j)_i^2$. Let U place the vectors u_j on disjoint coordinate blocks. Complete Q_0 to an orthogonal matrix $[Q_0 \ R_0]$, and let Δ contain the reserved weights. Set

$$Q' = UQ_0, \quad C' = \text{diag}(D_1, \dots, D_{2k}) + UR_0\Delta R_0^T U^T.$$

The two summands act on orthogonal subspaces. Thus C' has exactly the m' remaining positive weights and kernel range(Q'). There are $m' + k$ coordinates, as required. Every invertible k -row selection of Q' uses distinct groups; writing it as $Z(Q_0)_H$, where Z is diagonal and invertible, gives

$$C'_{I,I} \geq Z \text{diag}(v_j : j \in H) Z, \quad \text{tr}((Q'_I)^{-1} C'_{I,I} (Q'_I)^{-T}) \geq \frac{c_0}{2} \sqrt{k} s.$$

Append the first $8k$ weights as singleton coordinates, and zero rows to Q' , giving Q, C in dimension n . Diagonalize $C = RBR^T$ with $[Q \ R]$ orthogonal. Complementary bases Q_I, R_J are invertible together. If $E = R_I$, then

$$(I - E^T E)^{-1} = I + E^T (I - EE^T)^{-1} E,$$

and taking a B -weighted trace gives

$$\text{tr}(B(R_J^T R_J)^{-1}) = s + \text{tr}(Q_I^{-1} C_{I,I} Q_I^{-T}). \quad (\text{E.6})$$

Every invertible Q_I avoids the appended zero rows and is covered by the preceding estimate. This is stronger than (E.1) in the remaining diffuse case.

For the spectral transfer, complete R to $[Q \ R]$ and give Q the leading covariance eigenvalues. The resulting precision satisfies $K^{-1} \leq a^{-1} I + RB^{-1} R^T$. For an invertible complementary basis, the inverse eigenvalues of $R_J B^{-1} R_J^T$ have total at least $L = \gamma(k, m)s$. Scalar regularization therefore gives error at least $aL/(a + L)$. A singular basis contributes an inverse eigenvalue a after regularization and gives the same lower bound. The diagonal orientation also gives $x_k \geq s$. Combining these facts with $y_k \leq (k + 1)s$ proves (E.2).

Directly from the two cases defining γ , $(k + 1)/\gamma(k, m) \leq (16/c_0)\sqrt{r}$. This proves the finite trace-condition consequence. For fixed boundary coefficients, put $a_* = \min_i a_i$ and $s_* = \sum_j b_j$. For sufficiently small ϵ , the displayed a_i are the leading entries, and (E.2) gives

$$\frac{x_k(\lambda(\epsilon))}{\epsilon} \geq \frac{a_* \gamma(k, m) s_*}{a_* + \epsilon \gamma(k, m) s_*}.$$

Meanwhile direct product expansion gives $y_k(\lambda(\epsilon))/\epsilon \rightarrow (k + 1)s_*$. Dividing by the positive lower limit proves (E.3), without exchanging an orbit maximum with a limit. $\square$

Corollary E.2 (A finite balanced condition-number bound). *For $n = 2k$, put $a = \lambda_k$, $A = \lambda_1$, $s = \sum_{i>k} \lambda_i$, and $\kappa = A/a$. Then*

$$y_k \leq (k + 1) \min\{s, \sqrt{As/k}\}, \quad \frac{y_k}{x_k} \leq (k + 1) \min\left\{1, \frac{32 \cdot 10^9 + \sqrt{\kappa}}{\sqrt{k}}\right\}. \quad (\text{E.7})$$

For fixed arbitrary positive boundary coefficients in balanced dimension, the sharper bound in (E.3) is $32 \cdot 10^9(k + 1)/\sqrt{k}$.

Proof. The last assertion is the $m = k$ value of γ . To prove the additional finite estimate, the quotient $F = e_{k+1}/e_k$ is coordinatewise nondecreasing by (C.4). It is also nondecreasing when two entries of fixed sum are averaged. Indeed, writing $z = x + y$, $p = xy$, and q_j for the other entries' elementary coefficients, its derivative with respect to p has numerator

$$z(q_{k-1}^2 - q_{k-2}q_k) + (q_{k-1}q_k - q_{k-2}q_{k+1}) \geq 0.$$

The sign follows from the log-concavity and decreasing coefficient ratios proved in Lemma 8.1, with zero boundary coefficients interpreted as there. Thus raising the first k entries to A and averaging the last k to s/k cannot decrease F . For completeness, full averaging follows by maximizing F on the compact simplex of tail weights and choosing a maximizer of least sum of squares: averaging two unequal entries contradicts that choice. The denominator stays at least A^k even on the boundary.

Set $\epsilon = s/(kA) \leq 1$ and $\beta = (1 + \epsilon)/\sqrt{\epsilon} \geq 2$. The polynomial identity

$$(1 + \beta u + u^2)^k = \sum_{j=0}^k \binom{k}{j} (\beta - 2)^j u^j (1 + u)^{2(k-j)}$$

has nonnegative summands, in each of which the degree- $k + 1$ coefficient is no larger than the degree- k coefficient. Rescaling back shows $F(1^k, \epsilon^k) \leq \sqrt{\epsilon}$. Hence $F(\lambda) \leq \sqrt{As/k}$; the other bound $F \leq s$ was already proved.

Write $t = s/a$ and $d = 32 \cdot 10^9/\sqrt{k}$. Combining these estimates with (E.2) gives

$$\frac{y_k}{(k+1)x_k} \leq \min\{1, \sqrt{\kappa/(kt)}\} \min\{1, d+t\}.$$

For $t \leq 1$ the right side is at most $d+t \min\{1, \sqrt{\kappa/(kt)}\} \leq d + \sqrt{\kappa/k}$. For $t \geq 1$ it is at most $\sqrt{\kappa/k}$. It is always at most one, proving (E.7). $\square$

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